

Modeling Problems

I. Multicollinearity

Two or more predictors are highly/perfectly correlated.

$$X = \begin{pmatrix} & X_1 & X_2 & \cdots & X_{p-1} \\ 1 & X_{11} & X_{21} & \cdots & X_{(p-1)1} \\ | & X_{12} & X_{22} & & \vdots \\ \vdots & \vdots & \vdots & \ddots & \\ | & X_{1n} & X_{2n} & \cdots & X_{(p-1)n} \end{pmatrix}_{n \times p}$$

What if one of these
is linearly dependent (linear combo),
or close to it, on the others?

$$\begin{aligned} \text{Ex: } X_1 &= X_2 + X_3 \\ X_2 &= 5X_6 + 7X_7 \end{aligned}$$

$$\hat{\beta} = (\underline{X^T X})^{-1} X^T Y$$

might have issues

$$\text{Ex: } y \sim x_1 + x_2 \quad Y = X\beta + \varepsilon \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}$$

$$\text{but } x_2 = 4x_1$$

$$X = \begin{pmatrix} 1 & 1 & 1 \\ \vdots & x_1 & x_2 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\text{Suppose we know } \beta = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$$

SAME MODEL
DIFFERENT
 β 's

$$y_i = 1 + 2x_{1i} + 4x_{2i} + \varepsilon_i \quad \text{TRUE MODEL}$$

$$y_i = 1 + 2x_{1i} + 4(4x_{1i}) + \varepsilon_i$$

$$y_i = 1 + 18x_{1i} + 0x_{2i} + \varepsilon_i$$

$$\beta = \begin{pmatrix} 1 \\ 18 \\ 0 \end{pmatrix}$$

$$y_i = 1 + 22x_{1i} - x_{2i} + \varepsilon_i \quad \beta = \begin{pmatrix} 1 \\ 22 \\ -1 \end{pmatrix}$$

$$1 + 22x_{1i} - 4x_{1i} + \varepsilon_i$$

$$= 1 + 18x_{1i} + \varepsilon_i$$

WHICH ONE IS RIGHT?

Non-identifiability

- ① $\hat{\beta} = (X^T X)^{-1} X^T Y$ could be impossible to calculate
or very unstable because division
by ≈ 0

(2) $\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1} \leftarrow \text{explodes}$
 Inflated variance of $\hat{\beta}_s$

Inverse = $\frac{1}{\text{determinant}}$ x Mat

(3) Inference

$$t = \frac{\hat{\beta}_i - 0}{\underbrace{\widehat{SE}(\hat{\beta}_i)}_{\text{this is huge}}}$$

Always really small, never reject H_0 . Low power

Symptoms: adding predictor increase SE of others or change $\hat{\beta}_s$ by "a lot".

Ex:

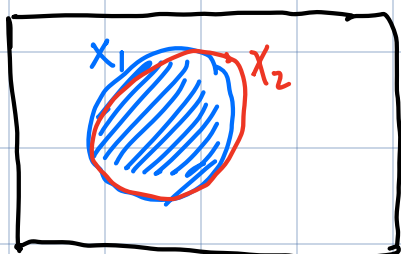
$\text{corr}(X_1, X_2) \approx 0$

Model	$\hat{\beta}_1$	$\hat{\beta}_2$
$y \sim X_1$	-1	
$y \sim X_1 + X_2$	-1.1	-5

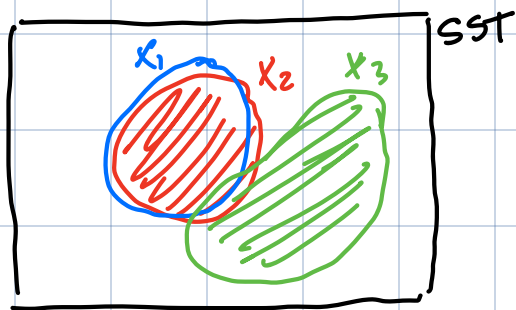
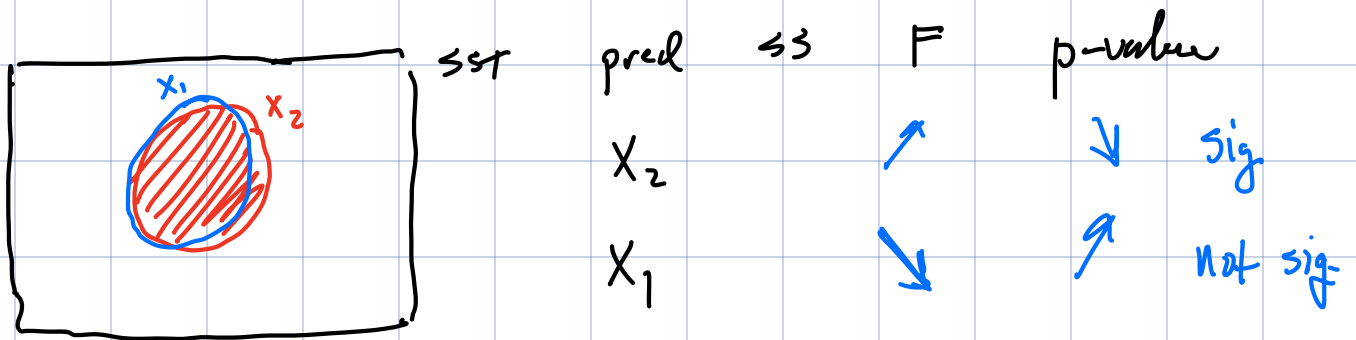
$\text{corr}(X_1, X_2) \approx 1$

$\hat{\beta}_1$	$\hat{\beta}_2$
-1	
10	-4

ANOVA



	pred	SS	F	p-value	
	X_1		$\uparrow$	$\downarrow$	signif.
	X_2		$\downarrow$	$\uparrow$	not sig.



Detection :

① Correlation Matrix

	X_1	X_2	X_3	X_4
X_1	1	.9	.2	⋮
X_2	.9	1	.3	⋮
X_3	.2	.3	1	
X_4		⋮		1

Drop one of X_1 or X_2

② Variance Inflation Factor (VIF)

VIF measures inflation of the variance of $\hat{\beta}_j$ when you add X_j to the model.

$$VIF = \frac{1}{1 - R_j^2} \quad \text{where } R_j^2 \text{ is } R^2 \text{ for } X_j \sim X_1 + X_2 + \dots + X_{j-1} + X_{j+1} + \dots + X_{p-1}$$

If R_j^2 is high, X_j is close to linear combo of other predictors, $VIF \uparrow$

If $VIF = 1$ no correlation with other predictors

$1 \leq VIF \leq 4$	light
$4 < VIF \leq 10$	moderate
$VIF > 10$	severe

Solutions:

- ① Drop correlated predictor, keep one
- ② Feature eng. $X^* = g(\text{correlated } X_s)$
- Intro ML — ③ Regularized regression LASSO, Ridge
- ④ Partial LS
- ⑤ PCA → adv ML

II Outliers + Influential Points

Leverage

$$\hat{Y} = H Y$$

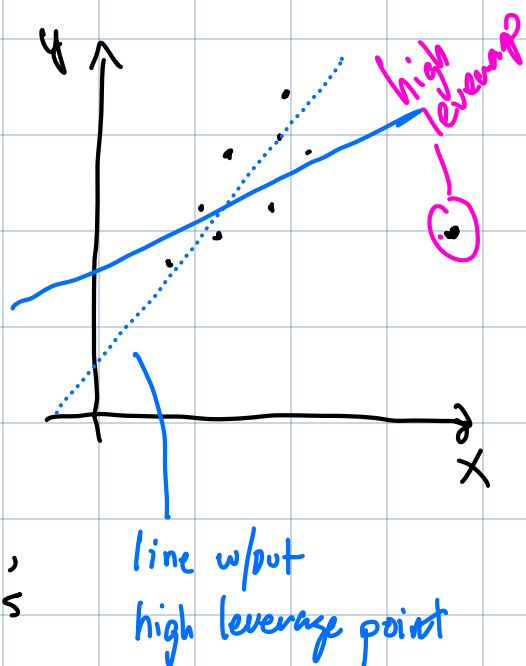
$n \times n$ $n \times 1$

$$H = \begin{pmatrix} h_1 & & \\ & \ddots & \\ & & h_n \end{pmatrix} \quad Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$\hat{y}_i = h_{i1} y_1 + h_{i2} y_2 + \dots + h_{in} y_n$$

h_{ii} = influence of y_i on $\hat{y}_i$

h_{ii} is the leverage of the i th point, and is a measure of distance between X_i 's at the i th point and average X 's



$$\text{In SLR } h_{ii} = \frac{1}{n} + \frac{(X_i - \bar{X})^2}{\sum (X_j - \bar{X})^2}$$

Properties of h_{ii}

(1) $0 \leq h_{ii} \leq 1$

(2) $\sum_i h_{ii} = p \quad \bar{h} = \frac{p}{n}$

High leverage $> 3 \frac{p}{n}$